2c. Content of Research Methods (Components 01 and 02)

There will be a research methods section within each component. For component 01, the focus will be on designing an investigation and for component 02, the assessment will relate to a novel source. Specific research methods content may also be assessed within the topic areas.

Learners are encouraged to carry out ethical, investigative practical activities appropriate for the study of psychology at this level, but learners will not be directly assessed on these specific activities.

A minimum of 20% of the overall marks in both components are for the assessment of Research Methods.

A minimum of 10% of the overall marks across the two components are targeted at the assessment of mathematics relevant to research methods in psychology (please see Section 5c).

<u>Planning Research</u>	Learners should have knowledge and understanding of the following features of planning research and their associated strengths and weaknesses, including reliability and validity.
Hypotheses	• Null and alternative hypotheses • Hypotheses to predict differences, correlations, or no patterns.
Variables	• Independent variables and how they can be manipulated • Dependent variables and how they can be measured • Co-variables and how they can be measured • Extraneous variables and how they can be controlled, including the use of standardisation.
Experimental Designs	• Repeated measures design • Independent measures design.
Populations and Sampling	• Target populations, sampling and sample size with reference to representativeness and generalisability • Sampling methods; random, opportunity, self-selected • Principles of sampling as applied to scientific data.
Ethical Guidelines	• Ethical issues: ◦ lack of informed consent ◦ protection of participants / psychological harm ◦ deception. • Ways of dealing with ethical issues: ◦ use of debriefing ◦ right to withdraw ◦ confidentiality. • The British Psychological Society’s Code of Ethics and Conduct.

Doing Research	Learners should have knowledge and understanding of the following features of doing research and their associated strengths and weaknesses including reliability and validity and the type of research objectives for which they are most suitable.
Experiments	• Laboratory • Field • Natural.
Interviews	• Structured • Unstructured.
Questionnaires (Surveys)	• Open questions • Closed questions • Rating scales.
Observations	• Naturalistic • Controlled • Overt • Covert • Participant • Non-participant.
Case Studies	• Use of qualitative data • Use of small samples.
Correlations	• Use of quantitative data • Positive, negative and zero correlations.

<u>Analysing Research</u>	Learners should be able to demonstrate knowledge and understanding of the process and procedures involved in the collection, construction, interpretation, analysis and representation of data. This will necessitate the ability to perform some calculations.
Types of Data	• Quantitative data • Qualitative data • Primary data • Secondary data • Strengths of each type of data.

Descriptive Statistics	• Measures of Central Tendency:
	    ○ Mode (including modal class)
	    ○ median
	    ○ mean.
	• Range
	• Ratios
	• Percentages
	• Fractions
	• Expressions in decimal and standard form
	• Decimal places and significant figures
	• Normal distributions
	• Estimations from data collected.

Tables, Charts and Graphs	• Frequency tables (tally chart) • Bar charts • Pie charts • Histograms • Line graphs • Scatter diagrams.

Reliability and Validity	• Reliability:
	    ○ internal
	    ○ external
	    ○ inter-rater.
	• Validity:
	    ○ ecological
	    ○ population
	    ○ construct.
	• Demand characteristics
	• Observer effect
	• Social desirability.

Sources of bias	• Gender bias • Cultural bias • Age bias • Experimenter bias • Observer bias • Bias in questioning.